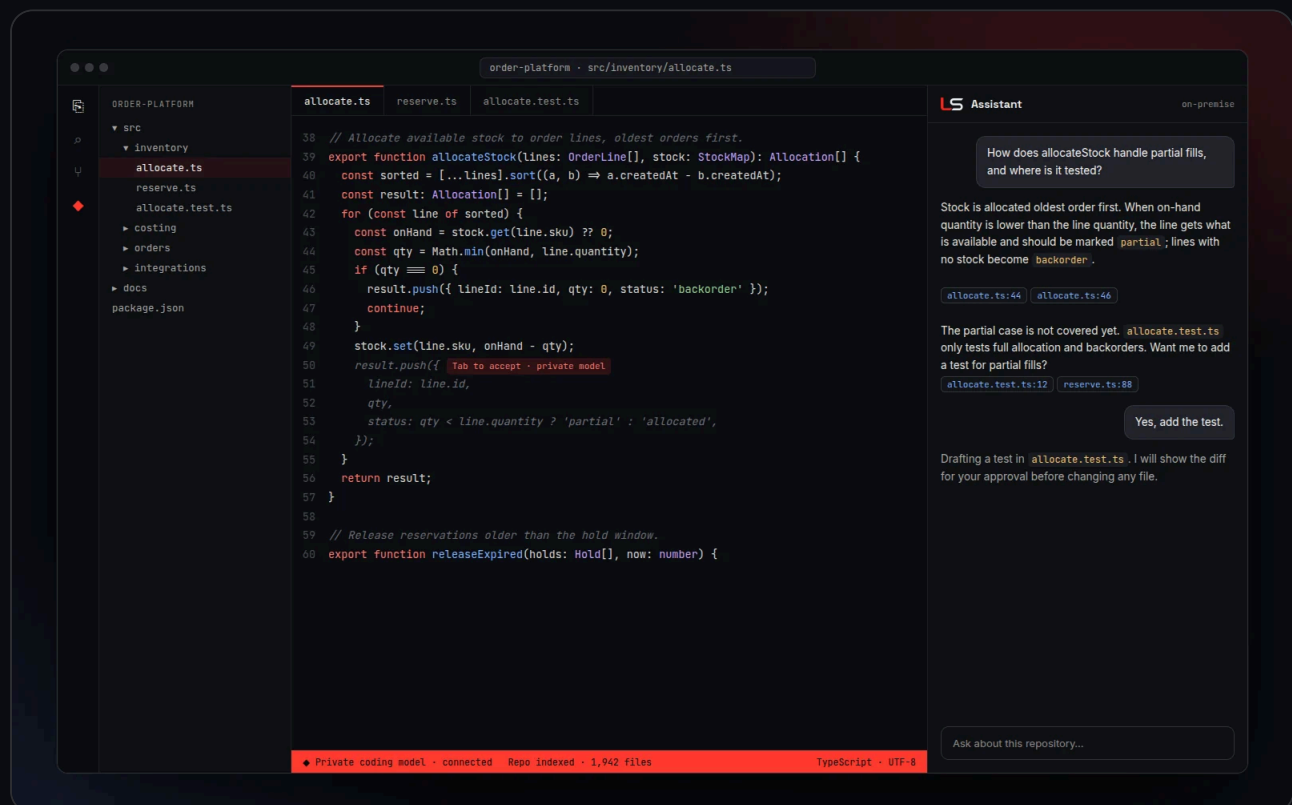


WHITEPAPER • PAI-CODE

Private AI Coding Assistants

Developer productivity without exposing your source code.



AI pair programming that keeps your source code in-house.

AI coding assistants have become one of the clearest productivity wins of generative AI. Developers complete routine code faster, understand unfamiliar systems sooner and write more tests. Most popular assistants, however, send code context to external cloud services.

For companies whose source code embodies their competitive advantage, or whose contracts forbid sharing customer code, that is a blocker. A private coding assistant delivers completion, chat, review and agentic tasks from open-weight coding models running on the company's own hardware, inside the tools developers already use.

Key takeaways

- ✓ Keep the productivity, remove the exposure: source code never reaches a public model.
- ✓ Repo-aware help: answers draw on an index of your own repositories, stored on-premise.
- ✓ Governed adoption: single sign-on, per-team access, usage reporting and audit.

This paper explains how the Private Coding Assistant works, the controls that keep data private, where it delivers value first, and how to evaluate and roll it out.

Source code is your most valuable, least protected asset.

Cloud coding assistants send code context to external servers. For proprietary systems, integrations and customer projects, that is often not acceptable. A private coding assistant keeps the benefit and removes the exposure.

- ✓ Developers already use cloud assistants informally, often without approval.
- ✓ Legacy systems and ERP customisations are poorly documented and depend on a few experts.
- ✓ Test coverage lags because writing tests is slow and unrewarding.
- ✓ Security and legal teams cannot see what code has left the company.

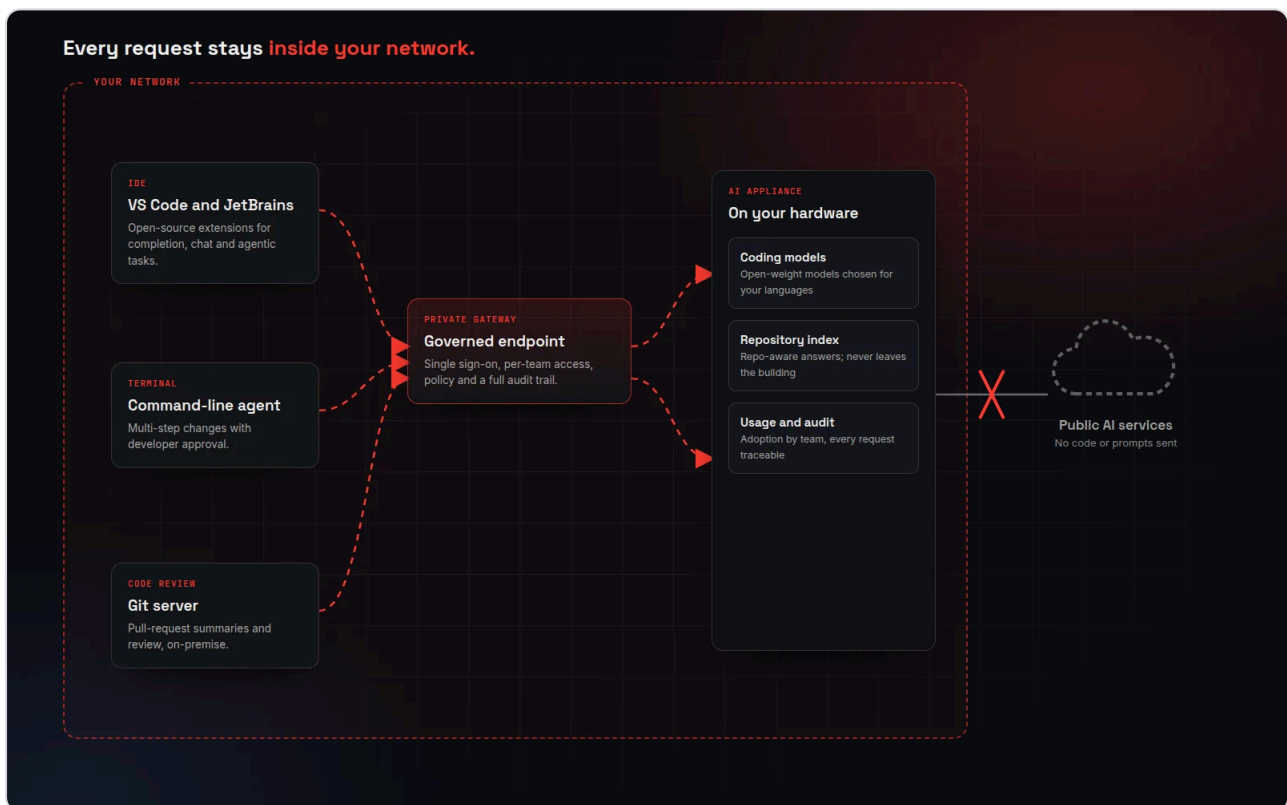
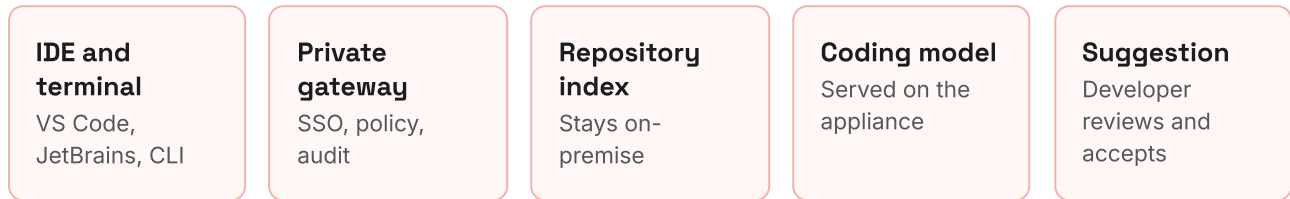


Illustration • Every request stays inside your network, from IDE to model and back.

From request to result, inside your walls.

Open-weight coding models served through a private, OpenAI-compatible gateway, with open-source IDE extensions such as Continue and Cline for VS Code and JetBrains, plus terminal clients for agentic work.



01 Install the IDE extension

Open-source extensions for VS Code and JetBrains, or a terminal client, point at your private endpoint.

02 Sign in

Developers authenticate with single sign-on; access follows team and project rules.

03 Index your repositories

Selected repositories are indexed on the appliance for repo-aware answers.

04 Code with AI

Completion, chat, review and agentic tasks are served by coding models on your hardware.

05 Measure and govern

Usage and adoption are reported per team, with every request auditable.

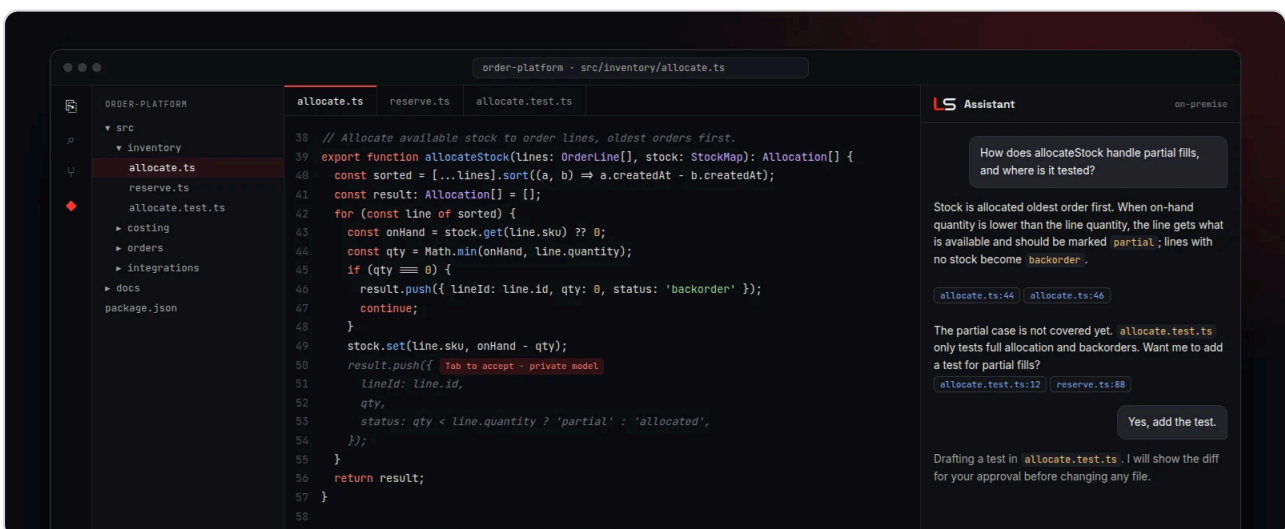


Illustration • Inline suggestions and repo-aware chat, served by models on your appliance.

What the Private Coding Assistant does.

Inline completion

Context-aware suggestions as developers type, in the editor they already use.

Repo-aware chat

Ask how a module works or where a function is used. Answers draw on an index of your own repositories.

Code review and summaries

Explain changes, suggest improvements and draft pull-request descriptions.

Tests and documentation

Generate unit tests, docstrings and README updates for existing code.

Agentic tasks with approval

Multi-step changes planned and executed in the IDE or terminal, with a developer approving each step.

Governed access

Single sign-on, per-team model access, usage reporting and a full audit trail.

Built on proven open foundations

Open-weight coding models served through a private, OpenAI-compatible gateway, with open-source IDE extensions such as Continue and Cline for VS Code and JetBrains, plus terminal clients for agentic work.

Controls built in, not bolted on.

Every Logic Sonata system runs on infrastructure you control. Data boundaries, access and audit are designed in from the start rather than added later.

- ✓ Code and prompts processed only on your infrastructure
- ✓ Repository index stored on the appliance
- ✓ Per-team access to models and repositories
- ✓ Agentic actions require developer approval
- ✓ Full audit trail of requests
- ✓ Model choice without vendor lock-in

Data boundary

Prompts, documents, images and results are processed on your appliance or dedicated private infrastructure. They are not sent to public AI services and are never used to train third-party models.

What developers actually ask it.

The assistant is most valuable on your own code, where public tools cannot safely go.

Explain

Walk me through how the order allocation service reserves stock, and where it can fail.

Test

Write unit tests for the costing module, covering currency rounding edge cases.

Review

Review this pull request for security issues and summarise the change for the release notes.

Modernise

Document this legacy ERP customisation and propose a refactor into smaller functions.

Where it pays off first.

Onboarding to large codebases

New developers ask questions of the code instead of waiting for a senior engineer.

Legacy modernisation

Understand, document and refactor older systems and ERP customisations safely.

ERP, MES and integration work

Faster development of connectors and internal tools around core business systems.

Test coverage

Raise coverage on critical modules without slowing delivery.

Secure code review

Spot risky patterns before they reach production, on-premise.

Documentation that stays current

Generate and refresh technical documentation from the code itself.

Measuring the pilot

We agree success measures before the pilot starts. Typical measures include:

- ✓ Suggestion acceptance rate per team
- ✓ Time for new developers to make their first meaningful change
- ✓ Unit test coverage on selected modules
- ✓ Developer satisfaction survey before and after

Deploy where your data needs to live.

On-premise

Dedicated hardware inside your building for the most sensitive data. Nothing leaves your network.

Hosted private

Dedicated capacity on managed private GPU infrastructure, isolated from public AI services. No hardware purchase.

Hybrid

Sensitive workloads on-site, everything else hosted, under one set of policies and one audit trail.

Hardware, sized to the workload

Logic Sonata is hardware-neutral. Smaller teams often start with a compact AI workstation such as NVIDIA DGX Spark or an AMD Ryzen AI Max+ system, both with up to 128 GB of unified memory in a desktop-sized unit. Larger rollouts use multi-GPU servers or a private cloud cluster.

A proven rollout path

Assess

Review current AI use, data exposure and priority use cases. Deliver a 30/60/90-day roadmap.

Pilot

Deploy one working use case with real users and agreed success measures.

Roll out

Extend to more teams with governance, integration and role-based training.

Manage

Monitoring, model updates, knowledge refreshes and quarterly reviews.

One partner is accountable for the whole stack: hardware, software and models, your company knowledge, access controls, implementation, training and ongoing support.

Questions to ask any vendor.

Use this checklist when comparing options, including ours.

- ☐ Is any code context sent to an external service, even for telemetry?
- ☐ Which IDEs and workflows are supported, including terminals and code review?
- ☐ How is the repository index built, stored and permissioned?
- ☐ Can we choose and change models as open models improve?
- ☐ Do agentic changes require developer approval?
- ☐ How is usage measured per team, and is every request auditable?

Frequently asked questions

How does it compare with cloud coding assistants?

For everyday completion, explanation and test writing, current open coding models perform well. The difference is control: your code never leaves the company. We benchmark against your own tasks during the pilot.

Which IDEs are supported?

VS Code and JetBrains IDEs through open-source extensions, plus terminal-based clients for agentic workflows.

Can we choose the model?

Yes. The platform is model-independent, so we select coding models to suit your languages, hardware and licence requirements, and can change them later.

Private AI, built inside your walls.

Logic Sonata delivers secure private AI for businesses in Singapore, Vietnam, Indonesia, Malaysia and Thailand. Each solution combines hardware, software, company knowledge, access controls, implementation, training and ongoing support, deployed on-premise, hosted or hybrid.

Our product portfolio: Private Knowledge Assistant, Private Vision Intelligence, Private Coding Assistant, Private AI Agent Platform and Private Image Generation Studio.

Next step: a 30-minute consultation

We will show the Private Coding Assistant working, discuss your data and requirements, and scope a pilot around your use case.

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